

FCOM audit fixes

Small upstream bugs found by the FCOM audit of every A32NX and A380X system (2026-10-02): eleven A380X fixes, six A32NX fixes, the fast ADIRS alignment of both aircraft and the A32NX ND chronometer colour, each checked against the FCOM, and an A32NX climb thrust regression test

Aircraft	A32NX and A380X
Where to find it	A380X: GPWS, OANS, SD (CRZ, PRESS, DOOR/OXY), FADEC, FWS (BRAKES HOT, T.O CONFIG, F/CTL ALTN LAW, HYD), ELEC panel, EWD EGT. A32NX: FWS (AIR PACK 2 OFF, ALL ENGINES FAILURE), BLEED, TCAS, SD (CRUISE, WHEEL), ND chronometer. Both: ADIRS
Branch	core/fcom-audit-fixes , based on master (FlyByWire master 255546cb0); independent of the other core branches. Each fix is small and can be taken upstream on its own. The twelfth A380X fix of the audit (SD FUEL jettison lines white during a normal jettison) needs the jettison of core/fuel and is in that branch.
Status	All fixes tested in the sim by the user on develop (2026-10-04). Not sim-tested on the master base; unit tests and Rust tests pass there.
Pictures	No pictures: every fix changes a stock FBW display, alert or system value.

1. A380X FCOM fixes
2. A32NX FCOM fixes
3. ADIRS fast alignment in 30 s
4. A32NX ND chronometer in white
5. A32NX climb thrust regression test

1. A380X FCOM fixes

Aircraft: A380X Where: GPWS, OANS, SD, FADEC, FWS, ELEC panel, EWD Status: Tested in the sim by the user on develop: OK (2026-10-04). Unit tests: LegacyGpws, OancSelectedRunway, FwsSystemDisplayLogicCrzPage, FadcWingAntilce, BrakesHot, ElecPanelPbs, EgtLimits, CabinAltitudeIndication, FwsAbnormalSensedItems, OxygenIndication specs; Rust test in hydraulic/mod.rs (each fails without its fix).
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What it is for

The FCOM audit of 2026-10-02 compared every A380X system with the A380 FCOM. These eleven findings were one or two wrong lines each (a variable nothing writes, a copy-paste index, an inverted condition). Each is now fixed as the FCOM describes.

What	Before	Now	FCOM
GPWS TOO LOW FLAPS	The FLAP MODE OFF selection was ignored (the GPWS read L:A32NX_GPWS_FLAP_OFF, which nothing writes)	TAWS FLAP MODE OFF on the MFD SURV CONTROLS page inhibits TOO LOW FLAPS	DSC-34-SURV
OANS selected runway	The FMS landing runway was compared with the FMS origin twice	The landing runway is green (selected) at the destination airport	DSC-34-10-70-20
SD automatic CRZ page	Engine 4 was not checked (engine 3 read twice)	Engine 4 alone at takeoff power also delays the CRZ page after takeoff	-
FADEC thrust limits	The wing anti-ice input read an A32NX variable that the A380X never writes	WING ANTI ICE ON (STRUCTURAL DEICE SWITCH) enters the thrust limit computation	DSC-70
BRAKES HOT and T.O CONFIG	Never triggered (L:A32NX_BRAKES_HOT is not written on the A380X)	From the 16 reported brake temperatures: one brake at or above 300 °C	PRO-ABN BRAKES HOT

ELEC panel COMMERCIAL 2 and ELMU pbs	All three pbs toggled COMMERCIAL (ELMU shed the commercial loads)	Each pb has its own state	DSC-24-20
EWD EGT amber limit	From the thrust lever angle	From the active thrust limit: none at FLEX/TOGA or in reverse, amber above the limit at MCT/CLB	DSC-70-90
SD PRESS CAB ALT needle	A320 delta P limits applied to feet	Red above 9 550 ft, like the value	DSC-21-30-20
F/CTL ALTN LAW procedure	MAX SPEED 310 KT shown only in ALT 1A (inverted)	Shown except in ALT 1A	PRO-ABN F/CTL ALTN LAW
HYD G RSVR LEVEL LO / G SYS OVHT	The G ELEC PMP A and B OFF line ticked from the yellow pumps; engine 3 and 4 pump disconnection on swapped DC buses	Green pumps; ENG 3 pump disconnection on DC 2, ENG 4 on DC 1	PRO-ABN HYD, DSC-29-40
SD DOOR/OXY crew supply	CREW SUPPLY OFF could not show; the CABIN line followed the crew oxygen pb	CKPT legend amber and REGUL PR LO on the ground with CREW SUPPLY OFF; the CABIN line amber on a low pressure only	DSC-35-20-20, DSC-35-30-20

Limitations and why

Limitation	Why	What to do
Only the wrong lines found by the audit were changed.	Each fix is kept to its finding so it can go upstream on its own; the rest of each system is as before.	-
ELMU and COMMERCIAL 2 have a state but no consumer yet.	Nothing in the electrical model sheds loads by those pbs (a380-simvars: A380X_OVHD_ELEC_COMMERCIAL_2_PB_IS_OFF, A380X_OVHD_ELEC_ELMU_PB_IS_OFF).	-
The EDP disconnection bus change has no cockpit test.	It only matters with DC 1 or DC 2 lost while disconnecting an engine pump; a Rust test covers it.	-

How to use

1. Nothing to do: the systems behave as in the FCOM.
2. TAWS FLAP MODE: MFD > SURV > CONTROLS, FLAP MODE OFF to inhibit TOO LOW FLAPS (e.g. a landing in a non-standard flap configuration).

How to test

#	Do this	Expected result
1	A380X on approach (FMS approach phase), gear down, flaps not in landing configuration, below 159 kt. MFD SURV CONTROLS: FLAP MODE on. Descend below 245 ft RA.	TOO LOW FLAPS.
2	Repeat with FLAP MODE OFF on the MFD SURV CONTROLS page.	No TOO LOW FLAPS.
3	Flight plan with a destination and a landing runway. At the destination, open the OANS map of the destination airport.	The FMS landing runway is shown green (selected runway).

4	After takeoff (climb), flaps and spoilers retracted, SD page not selected: keep thrust lever 4 alone at FLX/MCT or TOGA, levers 1-3 in CL.	The SD does not switch to the CRZ page at once: it waits for its 60 s timer, as with any engine at takeoff power. With all levers in CL it switches at once.
5	In flight, note the thrust limit on the E/WD; set WING ANTI ICE ON.	The thrust limit changes (the FADEC now sees the wing anti-ice).
6	After a high-energy stop (rejected takeoff or heavy braking), watch the SD WHEEL brake temperatures.	As soon as one brake reaches 300 °C: amber BRAKES HOT; the T.O CONFIG test then reports the hot brakes.
7	Overhead ELEC panel: press ELMU, then COMMERCIAL 2.	The COMMERCIAL pb stays as it was; each pb changes only itself.
8	Takeoff at FLX or TOGA, then thrust levers to CL.	EGT gauge: no amber EGT limit at FLX/TOGA; at CL the amber limit of the CLB thrust applies (EGT amber above it).
9	SD PRESS page with the cabin altitude above 9 550 ft (e.g. manual cabin altitude).	CAB ALT value and needle red.
10	Reach F/CTL ALTN LAW outside ALT 1A (e.g. with flyPad flight control failures).	The procedure shows MAX SPEED 310 KT; in ALT 1A it does not.
11	flyPad > Failures > ATA 29: activate Green reservoir leak ; when HYD G RSVR LEVEL LO shows, set the green electric pumps A and B OFF.	The procedure line of the green electric pumps A and B OFF ticks (the yellow pumps no longer tick it).
12	On the ground, overhead OXYGEN CREW SUPPLY pb OFF; SD DOOR/OXY page.	CKPT legend amber with REGUL PR LO; the CABIN line unchanged.

2. A32NX FCOM fixes

Aircraft: A32NX | **Where:** E/WD, SD BLEED, CRUISE and WHEEL pages, TCAS | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Unit tests: DeltaPIndication, AutoBrakeIndication, AllEnginesFailure, TcasGpwsInhibit specs; Rust test in pneumatic.rs (each fails without its fix). The OPT RELIGHT SPD clean-up (2026-10-05, a27ba6c35): AllEnginesFailure spec; the E/WD line shown is unchanged.

What it is for

The same audit on the A32NX found six one-line bugs. Each now follows the A320 FCOM. The ALL ENGINES FAILURE procedure was also cleaned up on 2026-10-05, with the LEAP-1A relight envelope of core/failures-a32nx.

What	Before	Now	FCOM
AIR PACK 2 OFF	Checked the ENG 1 BLEED pb	Checks the ENG 2 BLEED pb (each pack against its own bleed)	PRO-ABN AIR PACK 1(2) OFF
BLEED LO TEMP flag	Set only with the bleed valve closed (inverted)	Set when the bleed supplies air (valve open) below 150 °C in flight with WING ANTI ICE on; the bleed temperature turns amber on the SD BLEED page	DSC-36
TCAS TA ONLY on a GPWS alert	Read L:A32NX_GPWS_Warning_Active (an A380X variable)	A GPWS warning or alert (L:A32NX_GPWS_WARNING_LIGHT_ON / ALERT_LIGHT_ON) inhibits the RAs (TA ONLY)	DSC-34-SURV-60-10-10
SD CRUISE delta P pulse	Pulsed during the whole cruise	Pulses only in flight phase 7 above 1.5 PSI, resets at 1 PSI	DSC-21-20-40

SD WHEEL AUTO BRK colour	Amber with an engine off (single-engine taxi, engine failure)	Green when armed; amber only on an AUTO BRK failure (not modelled)	DSC-32-30-20
ENG ALL ENGINES FAILURE: APU START line	Only below 2 500 ft radio altitude	Below FL 250 (pressure altitude), text unchanged	PRO-ABN ENG ALL ENGINES FAILURE
ENG ALL ENGINES FAILURE: OPT RELIGHT SPD	Three unused lines (280/.77, 300/.77, 260/.77) in the message table next to the one shown; the line list written inline in the FWC	One line, OPT RELIGHT SPD.270/.77 (the line already shown, the LEAP-1A value); the line list and its conditions in FWC/Logic/AllEnginesFailure.ts with specs	Air Arabia A320 QRH ABN-19.01A ALL ENG FAIL (LEAP-1A; CFM56: 300/.77)

Limitations and why

Limitation	Why	What to do
The APU START line does not say "(BELOW FL 250)".	The 24-character ECAM line has no room; the condition gates the line instead, as for the other conditional lines.	-
AUTO BRK never turns amber.	The AUTO BRK failure that the FCOM links to the amber colour is not modelled.	-

How to use

1. Nothing to do: the alerts and pages behave as in the FCOM.

How to test

#	Do this	Expected result
1	A32NX in cruise. ENG 1 BLEED OFF, ENG 2 BLEED ON, X BLEED closed. Set PACK 2 pb OFF.	After 5 s: AIR PACK 2 OFF.
2	In flight with WING ANTI ICE ON, at low engine power (cold bleed air); SD BLEED page.	When a bleed supplies air below 150 °C, its bleed temperature is amber; with that ENG BLEED pb OFF (valve closed) it is not.
3	In flight with a TCAS RA possible (traffic), trigger a GPWS alert (e.g. SINK RATE).	The TCAS goes TA ONLY (no RA) while the GPWS alert is on.
4	SD CRUISE page in cruise with delta P above 1.5 PSI.	Delta P does not pulse. In the approach (flight phase 7) above 1.5 PSI it pulses, until it is below 1 PSI.
5	On the ground, one engine running, AUTO BRK MAX armed; SD WHEEL page.	AUTO BRK green (it was amber).
6	In flight above 2 500 ft RA and below FL 250, APU off: all engines flame out (e.g. fuel cut-off).	ENG ALL ENGINES FAILURE shows the APU START line, and one OPT RELIGHT SPD line: 270/.77.

3. ADIRS fast alignment in 30 s

Aircraft: A32NX and A380X | **Where:** Overhead ADIRS panel | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Rust tests in adirs.rs (the existing fast alignment test tightened to 30 s).

What it is for

A fast (re)alignment, the IR mode selector set to OFF and back to NAV within 5 s, took 180 s. Both FCOMs give about 30 s (A320 FCOM DSC-22_20-20-40, IRS alignment page: "a fast alignment takes 30 s"; A380 FCOM DSC-34-10-30-30). The shared systems code now takes 30 s.

Limitations and why

Limitation	Why	What to do
Only the fast alignment changed.	A normal alignment from an unaligned IR is unchanged.	-

How to use

1. With the IRs aligned on the ground, set an IR mode selector to OFF and back to NAV within 5 s.

How to test

#	Do this	Expected result
1	Aircraft on the ground, IRs aligned (NAV). Set IR 1 to OFF and back to NAV within 5 s; start a stopwatch.	IR 1 is aligned again after about 30 s (not 3 min).
2	Repeat on the A380X.	Same: about 30 s.

4. A32NX ND chronometer in white

Aircraft: A32NX | **Where:** ND, CHRONO pb on the EFIS control panel | **Status:** Tested in the sim by the user on develop: OK (2026-10-04). Spec ChronoColours.spec.ts (compiled styles of both aircraft).

What it is for

The ND chronometer was drawn in green digits on a light grey box, hard to read. The A320 FCOM gives "Chronometer Indication (white)" (DSC-31-45, INDICATIONS ON ND, item 9). The A32NX now shows white digits on a box in the display background colour.

Limitations and why

Limitation	Why	What to do
The dark box is a design choice.	The FCOM shows no box; a dark box keeps the digits readable over the map and terrain.	-
The A380X is unchanged.	Its FCOM gives no colour for the chronometer.	-

How to use

1. Press CHRONO on the EFIS control panel: the elapsed time appears at the bottom left of the ND. Press again to freeze it, a third time to reset and hide it.

How to test

#	Do this	Expected result
1	A32NX, ND powered: press CHRONO.	White digits on a black box at the bottom left of the ND, counting.
2	A380X: press CHRONO.	Unchanged: green digits on the grey box.

5. A32NX climb thrust regression test

Aircraft: A32NX | **Where:** Unit tests (FMS and FADEC climb thrust tables) | **Status:** Test only, no code change:
A320ClimbThrustLimit.spec.ts passes.

What it is for

A check of the A32NX climb against the A320neo FCOM climb tables (IndiGo FCOM PER-CLB-CLT, PDF pages 7248-7250 at ISA+10 and 7258-7260 at ISA+20, PW1127G-JM engines; no LEAP-1A table is available) found that the A32NX already climbs within 1-3 % of them, so no CLB thrust value was changed. The test keeps it so: the FADEC CLB N1 table (ThrustLimits_A32NX.hpp) and its FMS copy (cn1ClimbLimit) must stay identical row by row, and a climb computed with the FMS engine and flight models from FL 100 to FL 300 at 300 kt / M.78 must stay within 5 % of the FCOM times (8 % for each half) at 52 to 76 t.

Limitations and why

Limitation	Why	What to do
Nothing changes in the aircraft.	The finding was that the climb already matches the FCOM.	-
The reference is the PW1127G-JM table.	FBW models the LEAP-1A26; no LEAP climb table is available.	-

How to use

1. Nothing to do in the sim. When changing the CLB N1 table, change both copies and run the test.

How to test

#	Do this	Expected result
1	Run <code>npx vitest run fbw-a32nx/src/systems/fmgc/src/flightplanning/A320ClimbThrustLimit.spec.ts</code> .	All tests pass.
2	Optional in the sim: climbs at 52, 60 and 64 t, ISA+10, no wind, managed CLB speed, cruise FL 350; time from FL 100 to FL 300.	Within 5 % of 7.67 min (52 t), 9.08 min (60 t), 9.97 min (64 t); at ISA 2-5 % quicker.